Software Engineering
UA COLLEGE OF ENGINEERING

SFWE 407/507: Foundations of Software Engineering

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Course Overview

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Course Description

Software Engineering
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- This course will provide students with the foundational skills and knowledge used by software engineers in diverse industries.
- Students will understand what software engineering is and why it is more than *'just writing code'*.
- The course introduces students to the different software development lifecycle (SDLC) phases used in developing, delivering, and maintaining software products for a wide variety of applications.
- Common software process models will be introduced, along with developing an understanding of the importance of defining software requirements, developing software architectures and designs, and the various forms of testing that go into delivering reliable and resilient software systems.

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Learning Format

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- This course is designed to engage and demonstrate key concepts of the materials covered using collaborative learning strategies.
- Watch the pre-recorded lecture materials that may have interactive features integrated into the materials before coming to class. Tools such as Playposit, Perusall, and other software engineering specific tools may be used that allow you to demonstrate your understanding of the materials as you watch, read, and learn. During each module, the class will break into small teams or discussion groups to work on activities that demonstrate the key principles covered in the lectures.
- **Note:** Online students are not required to join the live sessions. I am going to record the live sessions.

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Course Map

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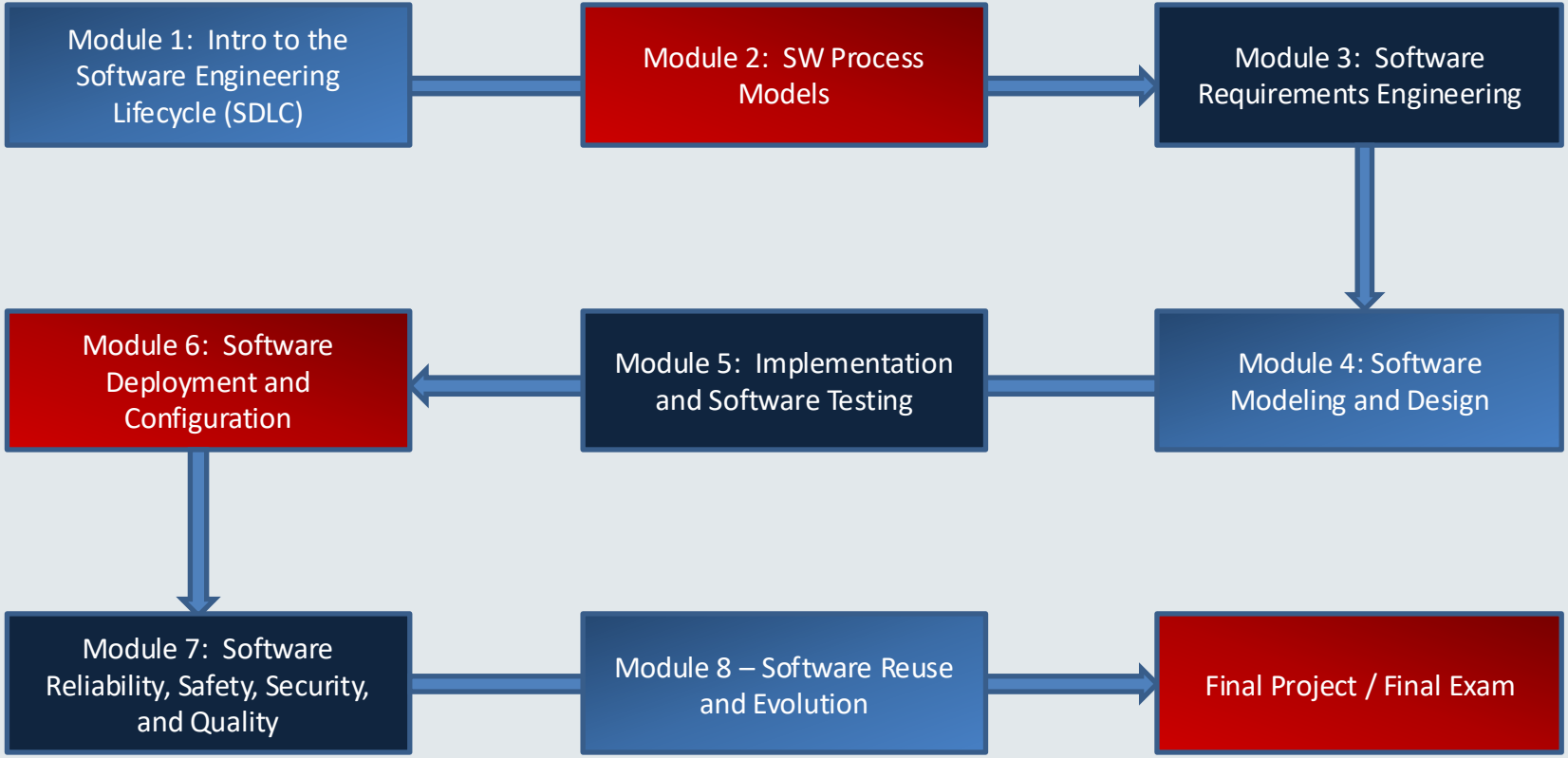

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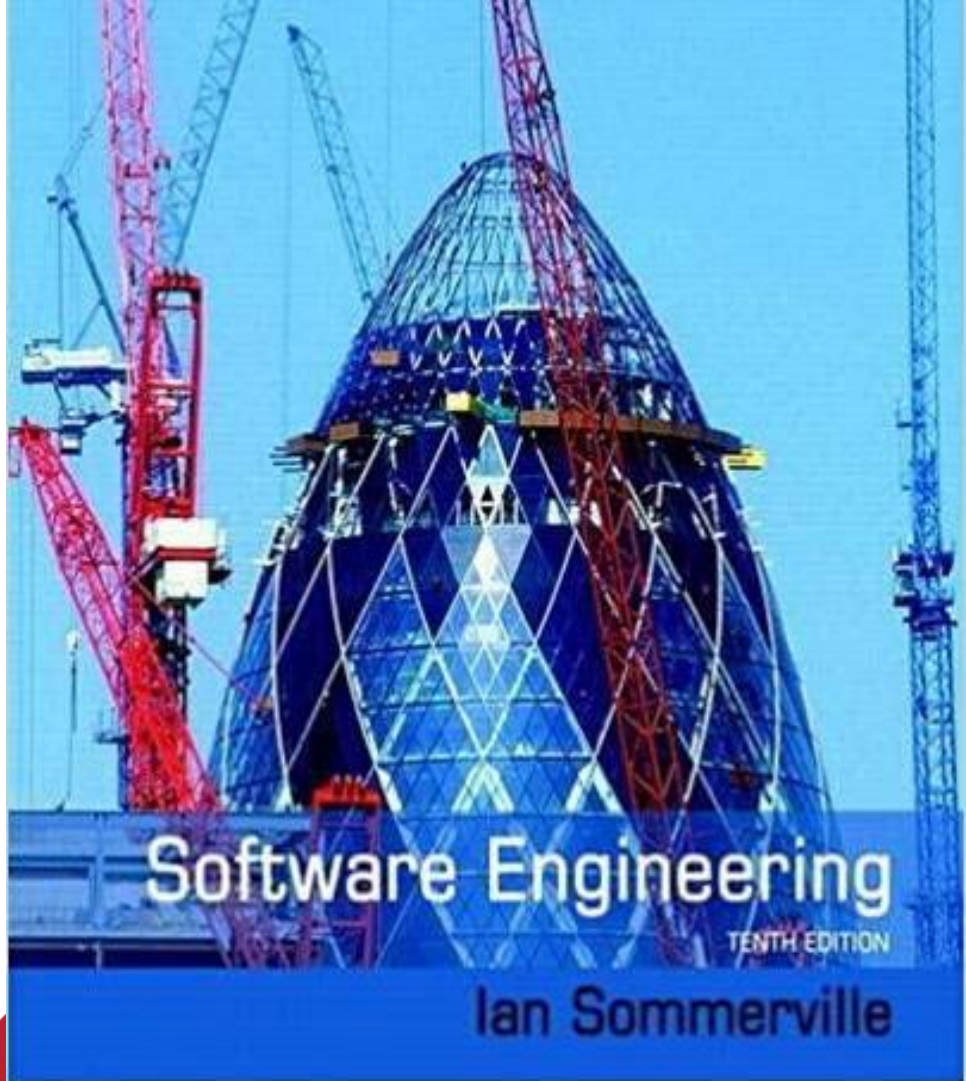
Textbooks

- Sommerville, Ian. Software Engineering, 10th Edition, Pearson Publishing, March 2015.

Other resources where you can find it:

1. https://www.amazon.com/Software-Engineering-10th-Ian-Sommerville/dp/0133943038
2. University Libraries: **O'Reilly Learning Platform (formerly Safari Books Online)**
3. Book available in the web: https://software-engineering-book.com/



Grading Scale and Policies

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- Homework Assignments: 20%
- Course Participation (weekly reflections): 10%
- Midterm Exam (1): 20%
- Comprehensive Final Exam (1): 20%
- Semester Project (1): 30%

- **NOTE:** As part of the semester project, graduate students will also be required to submit an additional paper that compares / contrasts the different software process models frequently used in industry. This paper will detail characteristics of each process model, the pros and cons of using the different models, and how using a different process model than what other engineering teams may be using on a large multi-disciplined project can impact the interactions with other teams.

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Due Dates / Late Work Policy

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➢ ALL DUE DATES ARE PUBLISHED ON D2L

- Check the due dates for all assignments and exams are published on the course calendar in D2L
- Homework is due at the time that it is specified in the course schedule and/or D2L content pages.
  - ***Late homework and projects <u>will not</u> be accepted without prior approval by the instructor and will receive 0 points.***
  - All dates and times mentioned in this course represent Mountain Standard Time (Arizona), which is UTC-7 hours.
    - Note: Arizona ***<u>does not</u>*** observe Daylight Savings Time.
  - A make-up exam may only be given in extreme circumstances and must have prior approval *(refer to syllabus)*

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What You Can Expect from your Instructor

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➢ WE ARE HERE TO SUPPORT AND FACILITATE YOUR LEARNING

- To always be treated with respect and dignity
- Regular announcements and important reminders posted on D2L
- Timely responses to your email / phone messages / in class questions
- Assistance with course materials / assignments / projects either during regularly scheduled office hours or at a time that is mutually convenient
- Rubrics that clearly outline the expectations for homework assignments and the semester project
- Constructive feedback on all assignments and projects

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Class Schedule (1/2)

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Module	Topic	Reading	Videos Lessons	Activities	Week	Module Begin / End		Date (in-person Class Meeting)	Deadlines and Dependencies*
						Begin	End		
START HERE Course Introduction	Course Introduction	Syllabus in the Start Here module on D2L	Start Here Video	Syllabus Challenge (D2L Quiz) Course Background Survey	1	14-Jan	18-Jan	No meeting/Complete the D2L Survey	
Semester Project Introduction	Semester Project Introduction	Semester Project Introduction module on D2L	Semester Project Introduction Video	Team introduction and selection	1	14-Jan	18-Jan	16-Jan	
MODULE 1 Introduction to the Software Engineering Lifecycle (SDLC)	Professional Software Development	Sommerville Chapter 1: 1.1	Professional Software Development	Micro Lab 1	2	14-Jan	25-Jan	21-Jan	No Class January 19th (MLK)
	Software Engineering Ethics	Sommerville Chapter 1: 1.2 GitHub Quickstart	Ethics in Software Engineering	Micro Lab 1	2	14-Jan	25-Jan	23-Jan	
	Case Studies	Sommerville Chapter 1: 1.3	Textbook Case Studies for Illustrating Concepts	Micro Lab 1	2	14-Jan	25-Jan	23-Jan	
	Micro Lab 1 | Introduction to Django, Frameworks, and Agile			**Team commitment, github repo and members, team roles, framework for project**	**1 and 2**				**Micro Lab 1 due 1/25/2026**
MODULE 2 Software Process Models	Introduction to Software Process Models and Process Activities	Sommerville Chapter 2: 2.1 and 2.2	Process Models Process Activities	Micro Lab 2	3	26-Jan	8-Feb	1/26 to 1/28	
	Managing Change and Continuous Process Improvement	Sommerville Chapter 2: 2.3 and 2.4	Managing Change Continuous Process Improvement	Micro Lab 2	3 and 4	26-Jan	8-Feb	1/30 to 2/2	
	Agile Software Development and Project Management	Sommerville Chapter 3: 3.1 to 3.4	Agile Software Development Agile Project Management	Micro Lab 2	4	26-Jan	8-Feb	2/4 to 2/6	
	Micro Lab 2 | Introduction to Django, Frameworks, and Agile			**Github and Team updates, language and framework for project, discussion and design for a Development Process**	**3 and 4**				**Micro Lab 2 due 2/8/2026**
MODULE 3 Software Requirements Engineering	Functional and Non-Functional Requirements	Sommerville Chapter 4: 4.1	Functional and Non-Functional	Micro Lab 3	4	9-Feb	22-Feb	2/9 to 2/11	No Class February 13th
	Requirements Engineering Process	Sommerville Chapter 4: 4.2 and 4.3	Requirements Engineering Process	Micro Lab 3	4 and 5	9-Feb	22-Feb	2/16 to 2/18	
	Requirements Specification, Validation, and Change Management	Sommerville Chapter 4: 4.4 through 4.6	Requirements Specification	Micro Lab 3	5	9-Feb	22-Feb	2/18 to 2/20	
	Micro Lab 3 | Defining Requirements for the Project			**Product Backlog, Sprint Planning, Demo**	**4 and 5**				**Micro Lab 3 due 2/22/2026**

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Class Schedule (2/2)

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MODULE 4 Software Modeling and Design	System Modeling	Sommerville Chapter 5: 5.1 to 5.5	System Modeling	Micro Lab 4	6	23-Feb	8-Mar	2/23 to 2/25	
	Architectural Design	Sommerville Chapter 6: 6.1 to 6.4	Architecture and Design	Micro Lab 4	6 and 7	23-Feb	8-Mar	2/27 to 3/2	
	Midterm Exam Guidance	D2L Module Exam | Midterm Exam	D2L Module Exam | Midterm Exam	Micro Lab 4	7	23-Feb	8-Mar	3/4 to 3/6	Spring Break March 7-15
	Micro Lab 4 | Product Backlog, Sprint Planning, Demo			Product Backlog, Sprint Planning, Demo	6 and 7				Micro Lab 4 due 3/16/2026 (Delay due to Spring Recess)
MIDTERM EXAM									
Module 5 Implementation and Software Testing	Design and Implementation	Sommerville Chapter 7: 7.1 to 7.4	Design and Implementation	Micro Lab 5	9	16-Mar	29-Mar	3/16 to 3/18	
	Software Testing	Sommerville Chapter 8: 8.1 to 8.4	Software Testing	Micro Lab 5	10	16-Mar	29-Mar	3/20 to 3/27	
	Micro Lab 5 | Product Backlog, Sprint Planning, Demo			Product Backlog, Sprint Planning, Demo	9 and 10				Micro Lab 5 due 3/29/2026
MODULE 6 Software Deployment and Configuration Management	Version Management and System Building	Sommerville Chapter 25: 25.1 to 25.2	Version Management and System Building	Micro Lab 6	11	30-Mar	12-Apr	3/30 to 4/3	
	Change and Release Management	Sommerville Chapter 25: 25.3 to 25.4	Change and Release Management	Micro Lab 6	12	30-Mar	12-Apr	4/6 to 4/10	
	Micro Lab 6 | Product Backlog, Sprint Planning, Demo			Product Backlog, Sprint Planning, Demo	11 and 12				Micro Lab 6 due 4/12/2026
MODULE 7 Software Reliability, Safety, Security, and Quality Assurance	Dependability and Reliability	Sommerville Chapters 10 and 11	Dependability and Reliability	Micro Lab 7	13	13-Apr	26-Apr	4/13 to 4/17	
	Safety and Security Engineering	Sommerville Chapters 12 and 13	Safety and Security Engineering	Micro Lab 7	13 and 14	13-Apr	26-Apr	4/17 to 4/22	
	Resilience Engineering	Sommerville Chapter 14: 14.1 to 14.3	Resilience Engineering	Micro Lab 7	14	13-Apr	26-Apr	4/22 to 4/24	
	MicroLab 7 Topic: Project Wrap-up for Showcase and Evaluation			Project Wrap-up for Showcase and Evaluation	13 and 14				Micro Lab 7 due 4/26/2026
MODULE 8 Software Reuse and Evolution	Software Reuse	Sommerville Chapter 15: 15.1 to 15.4	Software Reuse	Project Showcases and Evaluation Micro Lab 8	15	27-Apr	6-May	4/27 to 5/6	
	Software Evolution	Sommerville Chapter 9: 9.1 to 9.3	Software Evolution	Project Showcases and Evaluation Micro Lab 8	16	27-Apr	6-May	4/27 to 5/6	
	MicroLab 8 Topic: Product Reuse and Evolution			Product Reuse and Evolution	15 and 16				Micro Lab 8 (Individual, Final Exam Question)
Semester Project	Presentations and Peer Reviews			Presentations and Peer Reviews	15	27-Apr	6-May	4/27 to 5/6	
Semester Review	Discussion				16	27-Apr	6-May	4/27 to 5/6	8-May
FINAL EXAM									Monday May 11 10:30 AM to 12:30 PM

Meet Your Professor -

Dr Diana Saldana

dianasaldana@email.arizona.edu

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- **Experience:**
- In 2013 I started as Professor of Practice at UABC in the Computer Engineer program.
- In 2017, I moved to Yuma, AZ and I worked as an Engineering Professor at Arizona Western College.
- In 2019, I started working in the University of Arizona as Professor of Practice in the College of Applied Science and Technology where I taught Applied Computing and Computer Science courses.
- Moved to teaching Software Engineering courses in the College of Engineering at the University of Arizona in Fall 2022.
- **Education:**
- BS degree in Computer Engineering.
- MS and PhD in Computer Science.
- **Fun Facts:**
- I am *Swiftie* and a *Potterhead*.
- Tea Lover (especially boba tea <3)
- In my spare time I like to read novels and watch k-dramas!

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Quick introduction (please tell me your name and your major or if you are a graduate student)

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Let’s talk about the Semester Project

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Executive Summary

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- The Product Development Management System is an Agile project management tool that allows users to manage Product Backlog. This system allows team members to add new tasks to the Product Backlog, set and change priority for a task, move a task from the Product Backlog to the current Sprint Backlog, and then move a task from the Sprint Backlog to the Ready for Test status. Once tested, the task is either failed and sent back for re-work in a Sprint or passed and marked as complete and ready for release.

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Project Development

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- The release of this software is limited to prototyping in an iterative and incremental environment. The team will demonstrate partially running components of the software to show progress and to elicit comments, questions, and change requests after key Sprints.
- Expect and prepare for re-work. Sometimes you will find that what worked before adding a new feature is broken in the current demo release. This is exactly what happens in professional software development and is why this makes for a solid learning experience.
- We will be gathering and refining requirements, breaking them into manageable tasks, and using our own system to manage the building of this tool, incrementally, as each piece of functionality is added.

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Microlabs

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- We will use Micro Labs to progressively work on our project. This work may involve planning artifacts, user-interface mock-ups, working demos, and communications using presentations.

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Team Creation

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- Let's make teams with 4-5 members (You can submit your team's organization in MicroLab 1).
- You can combine in-person and online students or undergraduate and graduate students in your team.
- If you can't find a team, please let me know by Wednesday January 21st.

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Project Overview

https://d2l.arizona.edu/d2l/le/content/1717961/viewContent/20074267/View

Cover Letter

- Write a one-page cover letter for the Job Description posted in D2L:
- https://d2l.arizona.edu/d2l/le/1717961/discussions/List

SDLC Quick Refresher

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Requirements Analysis is Traditionally Performed Early in the SDLC ….but….(yes, there is ‘a but’)

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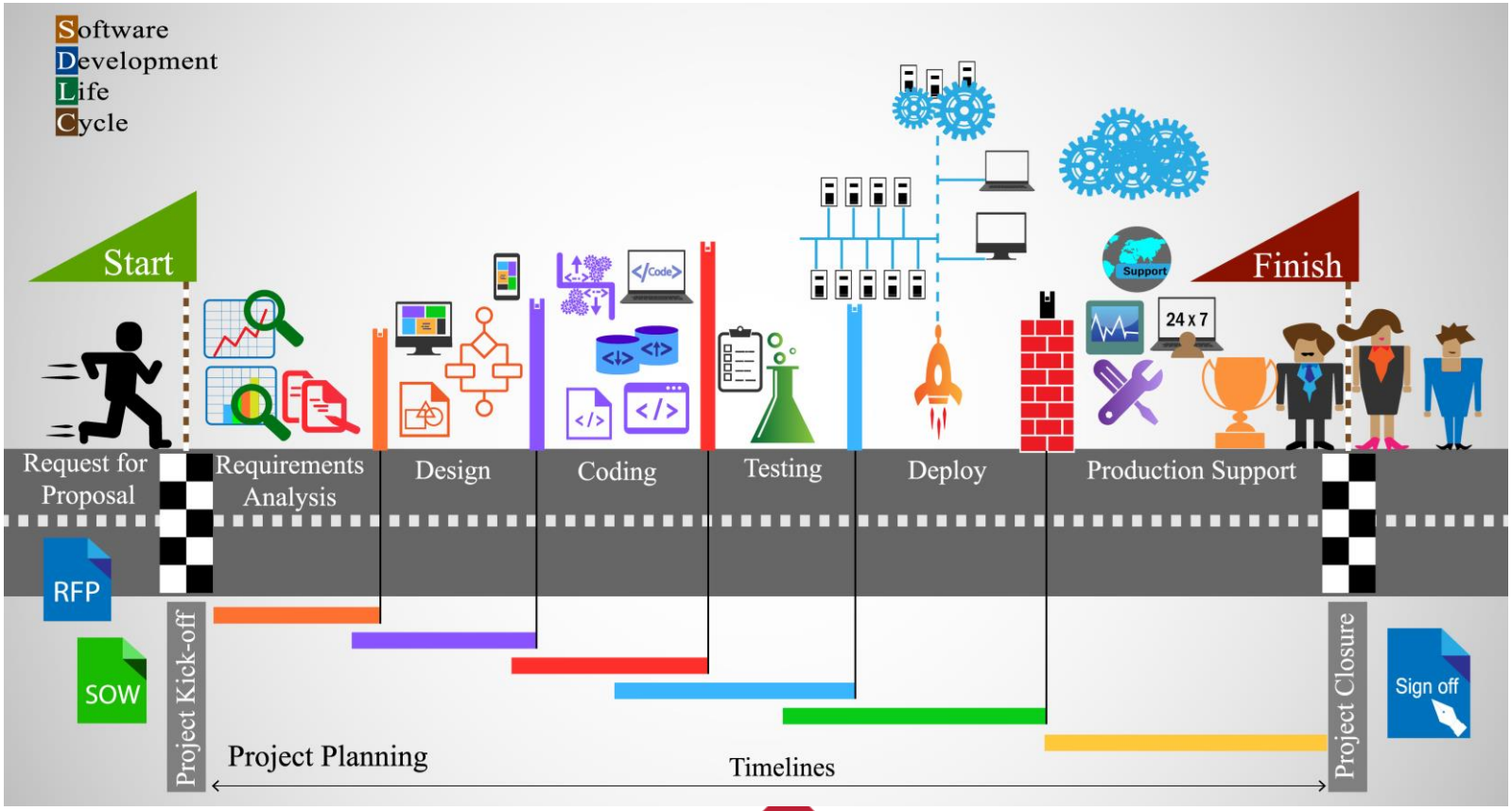

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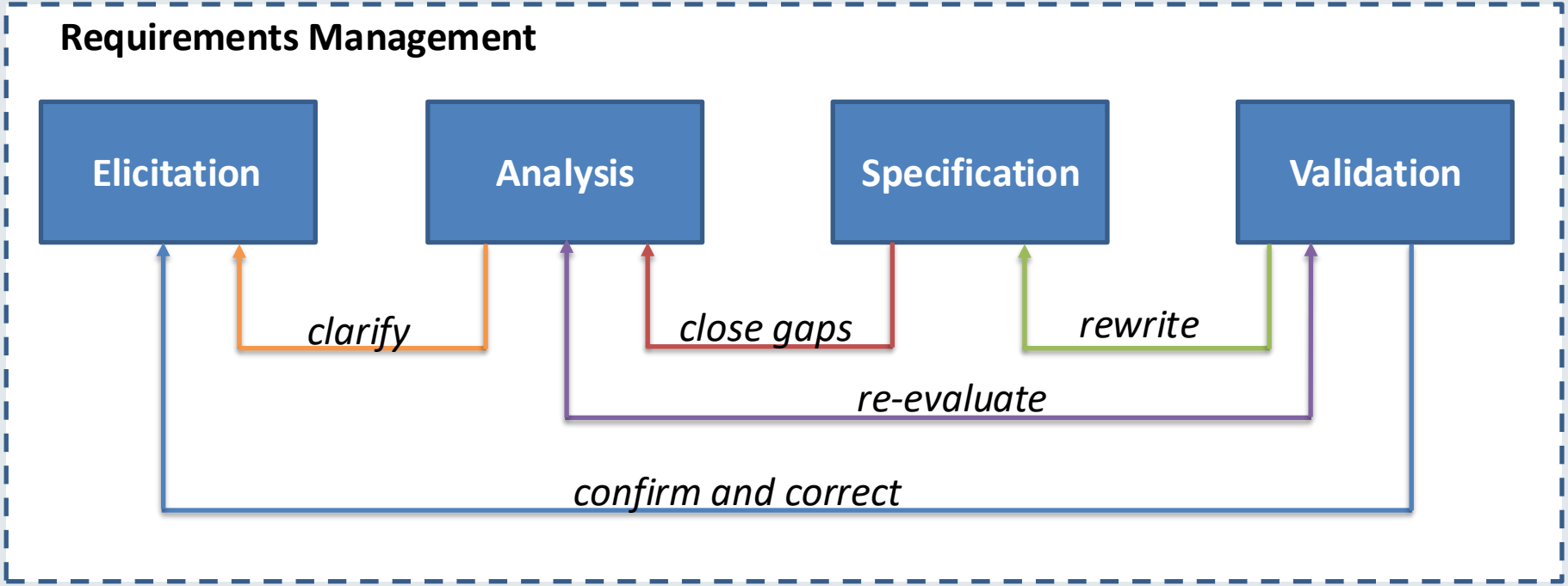
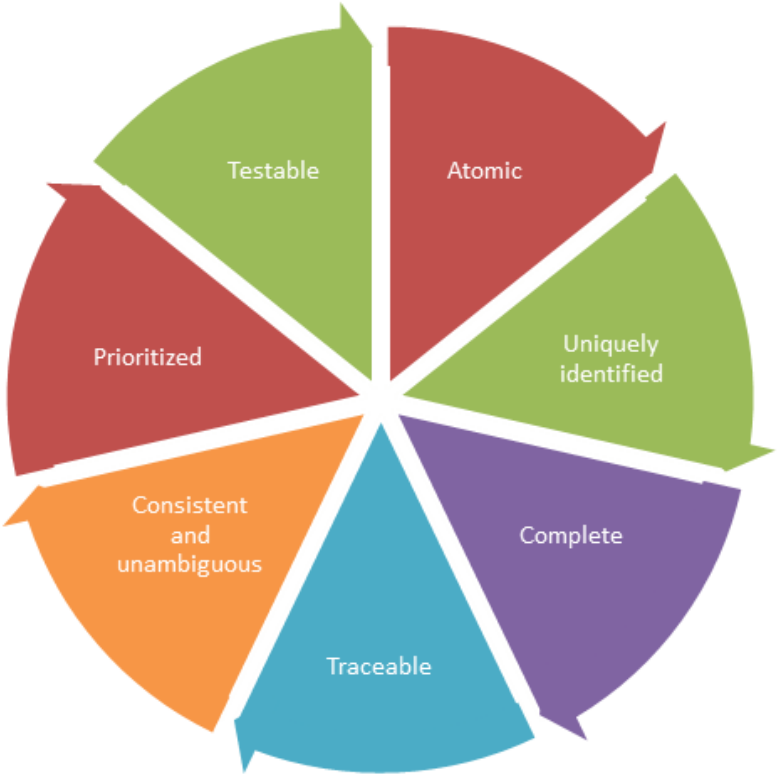
Requirements Engineering is an Iterative Process



What Constitutes a “Good” Software Requirement?

➢ ***GOOD, BAD AND THE UGLY***

- Software requirements should adhere to several standard qualities:
  - Atomic *(can it stand on its own?)*
  - Traceable (to higher level requirements and lower-level artifacts)
  - Uniquely Identified (facilitates traceability)
  - Complete (mitigate unnecessary work)
  - Consistent and Unambiguous (clear)
  - Prioritized (what should be developed first)
  - Testable (can it be verified?)



Simplified Software Requirements Engineering Process

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Start Here

Understand Business Requirements

Define User Requirements

Develop System Requirements

Flow System Requirements to Software (maintaining traceability)

Develop SW Functional Requirement

Develop SW Non-Functional Requirements

Evaluate "goodness" of the SW Requirement

Refine as Required

Establish a SW Requirements Baseline – Software Requirements Specification (SRS)

Manage Changes to the SW Baseline / SRS

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Overcoming Challenges with Traditional Requirements Engineering

Typical challenges

- Natural language problems (i.e., ambiguity and imprecision)
- Domain understanding
- Dealing with complexity
- Difficulties in capturing/defining system behavior
- Incompleteness
- Over specifying (too much design detail)
- Gold plating
- Over extension (i.e., dangerous “All”)
- Inconsistency
- Incorrectness
- Cost realism

Let’s discuss…

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SDLC philosophies

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Station	Methodology	Core Philosophy
A	**Waterfall**	Measure twice, cut once. Documentation is king.
B	**Agile (Scrum)**	Pivot quickly. Working software over documentation.
C	**V-Model**	Safety and Verification. Every stage has a mirror test.
D	**Spiral**	Risk-driven. If it might fail, prototype it first.
E	**Kanban**	Continuous flow. No deadlines, just constant throughput.

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Round 1: The “Mission Critical” Scenario

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- **Scenario:** You are building the flight control software for a new commercial spacecraft. A software bug could result in the loss of life and billions of dollars. Requirements are strictly defined by international physics standards and do not change.
- What SDLC philosophy or methodology you would choose?
- Why?

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Round 2: The “Market Disrupter”

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- **Scenario:** "You are a 3-person startup building 'Uber for Laundry.' You have $50,000 in the bank (enough for 3 months). You don't actually know if people want this service, and you need to show something to investors in 30 days.”
- What SDLC philosophy or methodology you would choose?
- Why?

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Round 3: The “Legacy Leviathan”

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- **Scenario:** "You are tasked with migrating a government Social Security database (40 years of data) to a modern cloud system. The project will take 4 years. Security is paramount, but the technology stack is so old that no one is 100% sure how the migration will affect the data.”
- What SDLC philosophy or methodology you would choose?
- Why?

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Round 4: The “Internal Support”

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- Scenario: "You are the IT department for a large law firm. You handle bug fixes, small feature requests (like 'add a logo to this report'), and password resets. There is no 'end date' for your work; it just keeps coming.”
- What SDLC philosophy or methodology you would choose?
- Why?

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